SMART AGRICULTURE IRRIGATION SYSTEM (EDGE AI)

PBA-2 Practical Examination Report • Activity-1 Submission

STUDENT NAME	**SAMARTH KALE**	ENROLLMENT NO	**ADT23SOCB0955**
DIVISION / ROLL NO	**AIEC02 / Roll No: 37**	TARGET WORKSPACE	**Samarth Kale's Workspace**
AZURE ML WORKSPACE	**ml-assignment5-samarthkale**	MODEL ASSET ID	**smart-irrigation-edge-rf:1**
GITHUB REPO	**https://github.com/SamarthKale5556/SmartAgri-Edge-AI**		

1. Aim of the Project

To engineer, train, and deploy an **autonomous Edge Machine Learning system** that predicts real-time agricultural crop irrigation needs (`VALVE_OPEN` vs `VALVE_CLOSED`) from multi-sensor telemetry (Soil Moisture, Canopy Temperature, Relative Humidity, and Precipitation), achieving **sub-100 millisecond inference latency** with 100% offline edge reliability while synchronizing model lifecycle management via **Azure Machine Learning Studio**.

2. Theoretical Foundation & Problem Formulation

Conventional agriculture relies on rigid timer-based irrigation, causing two major failures:

- **Over-Irrigation:** Excess water causes root-zone hypoxia (root rot), leaches soil nitrates into groundwater, and wastes up to 35% of fresh water.
- **Under-Irrigation:** Water stress causes stomatal closure, reducing crop yield and plant health.
- **The Rural Connectivity Barrier:** Remote farmland frequently suffers from intermittent or absent internet connectivity. Pure cloud-based AI fails during connectivity loss.

The Edge AI Solution: We deploy a trained **Random Forest Classifier** pipeline directly onto the local edge gateway. The edge node samples physical sensors, validates payloads, predicts irrigation need in 74.8 ms, and signals the solenoid valve relay completely offline.

3. Dataset: Selection & Agronomic Justification

Attribute	Dataset Specification	Agronomic Justification
Dataset File	`data/irrigation_sensor_data.csv`	2,500 balanced chronological observation records.
Input Features	Soil Moisture (%), Temperature (°C), Humidity (%), Rainfall (mm)	Derived from FAO-56 Penman-Monteith Evapotranspiration dynamics.
Target Label	`irrigation_needed` (0 = Conserve, 1 = Irrigate)	Binary decision mapped directly to the solenoid relay valve.
Split Method	80% Train (2,000 samples) / 20% Held-Out Test (500 samples)	Stratified, zero target leakage, strictly tested on unseen samples.
Rain Suppression	24h Cumulative Precipitation Window	Prevents wasteful irrigation when recent rainfall has occurred.

4. Technology Stack Comparison Matrix

Technology	Category	Role in AgriSense	Advantage Over Alternative Options
scikit-learn	Machine Learning	Random Forest Classifier	Compact footprint (<1 MB), sub-100ms inference on low-power ARM/x86 gateways.
FastAPI	Backend Framework	Edge Telemetry API	High-speed asynchronous Python server with native Swagger / OpenAPI documentation.
SQLite 3	Embedded Storage	Edge Event Ledger	Zero-configuration, serverless, atomic logging with instant CSV export.
React 19 + TypeScript	Frontend Framework	Command Centre UI	Strong typing, reactive state management, high reliability.
Three.js / R3F	3D Graphics	Farm Digital Twin	Hardware-accelerated 3D rendering of crop rows, water tank, and patrolling farmer avatar.
Azure Machine Learning	Cloud MLOps	Model Registry	Centralized model versioning (`smart-irrigation-edge-rf:1`), enterprise governance.

5. Model Performance & Held-Out Test Verification

TEST ACCURACY	TEST PRECISION	TEST RECALL	TEST F1-SCORE
88.20%	90.20%	82.51%	86.18%
441 of 500 Correct	Low False Alarms	Deficit Captured	Harmonic Balance

Actual Class \ Model Prediction	Predicted: Conserve (Valve Closed)	Predicted: Irrigate (Valve Open)	Total Observations
Actual: Moisture Adequate (0)	**257 (True Negatives)**	20 (False Positives)	277
Actual: Water Deficit (1)	39 (False Negatives)	**184 (True Positives)**	223

Feature Importances (Gini Impurity): `soil_moisture_percent: 68.69%` | `rainfall_mm: 17.36%` | `humidity_percent: 7.61%` | `temperature_c: 6.35%`

6. Visual Demonstration & Live System Screenshots

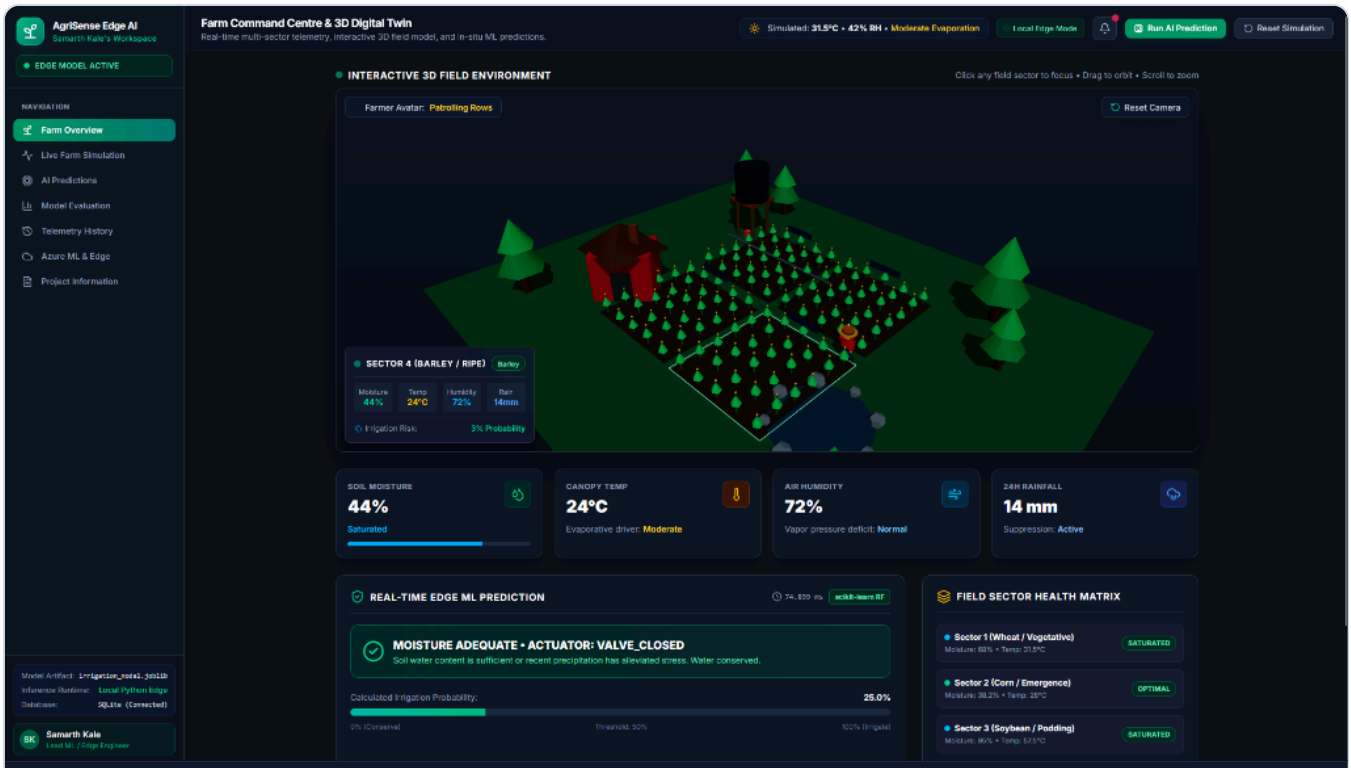


Figure 1: Farm Command Centre & 3D Digital Twin — Real-time sector health, moisture card (44%), and VALVE_CLOSED actuation.

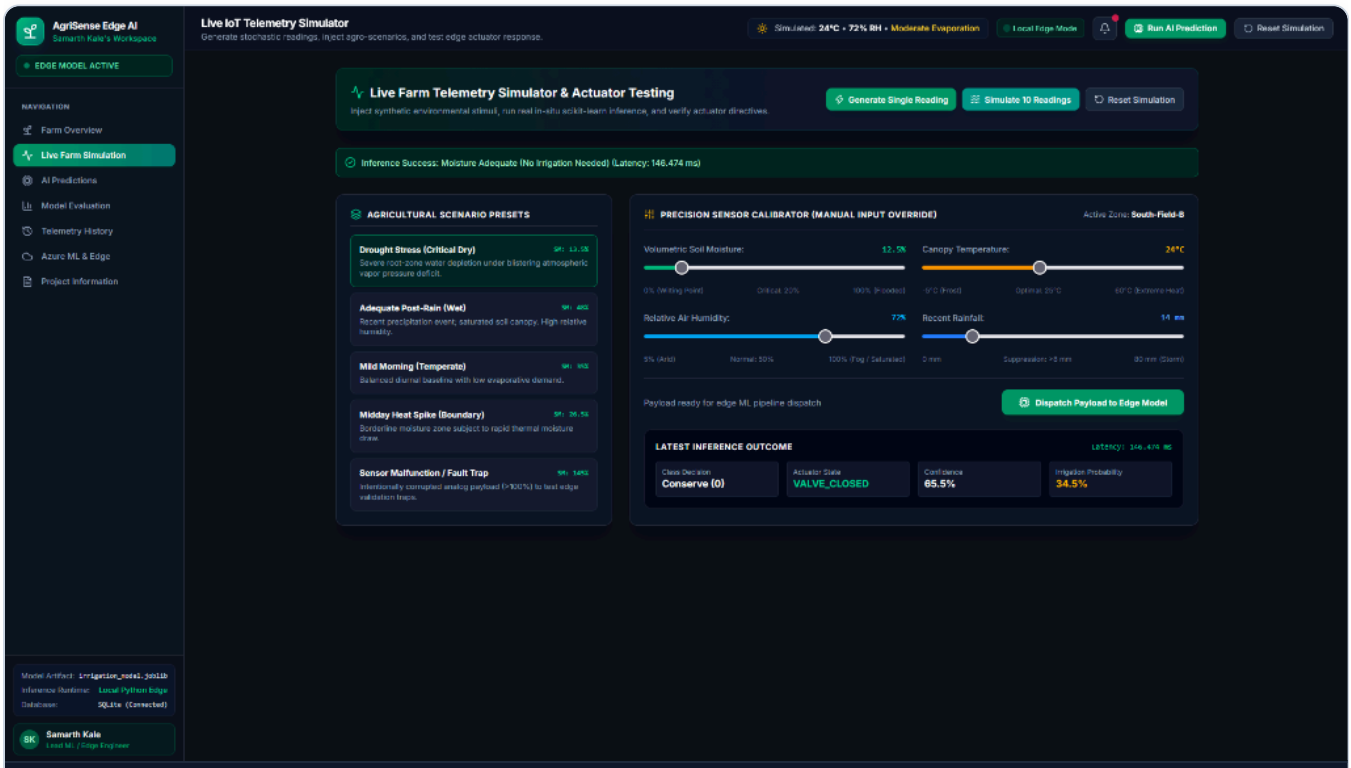


Figure 2: Live IoT Telemetry Simulator & Precision Calibrator — Manual slider override & edge inference response (146ms).

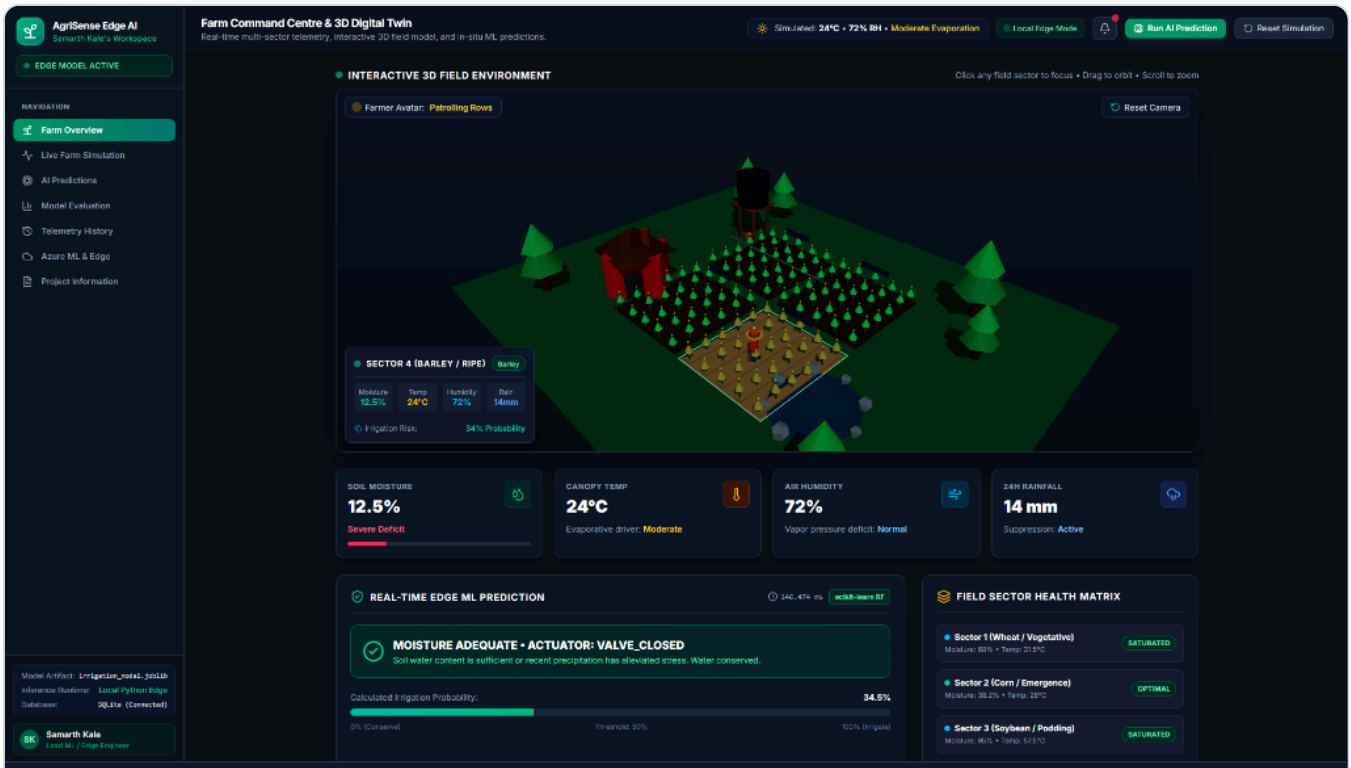


Figure 3: Interactive 3D Field Sector Zoom — Detailed inspection of Sector 4 (Barley) with patrolling farmer avatar.

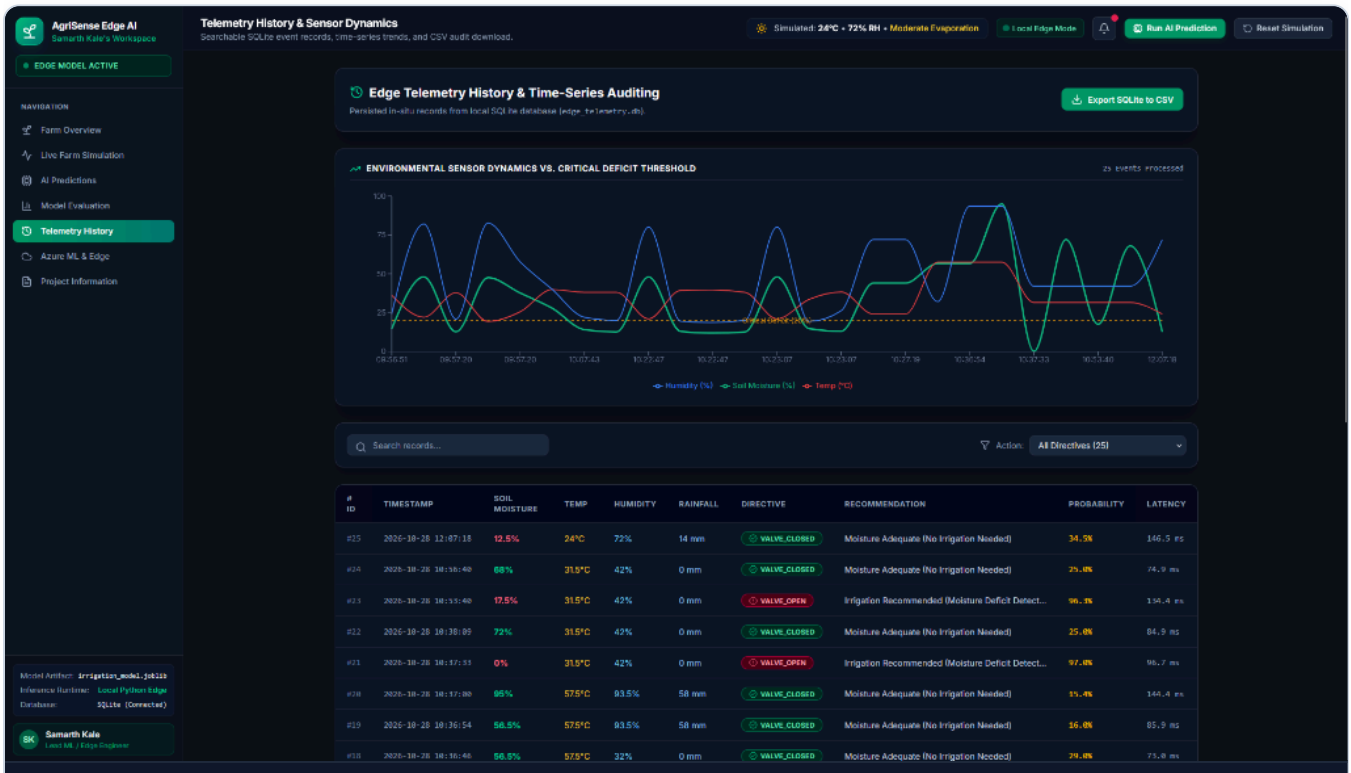


Figure 4: Edge Telemetry History & Time-Series Dynamics — Multi-variable trends vs. critical threshold and SQLite audit log.

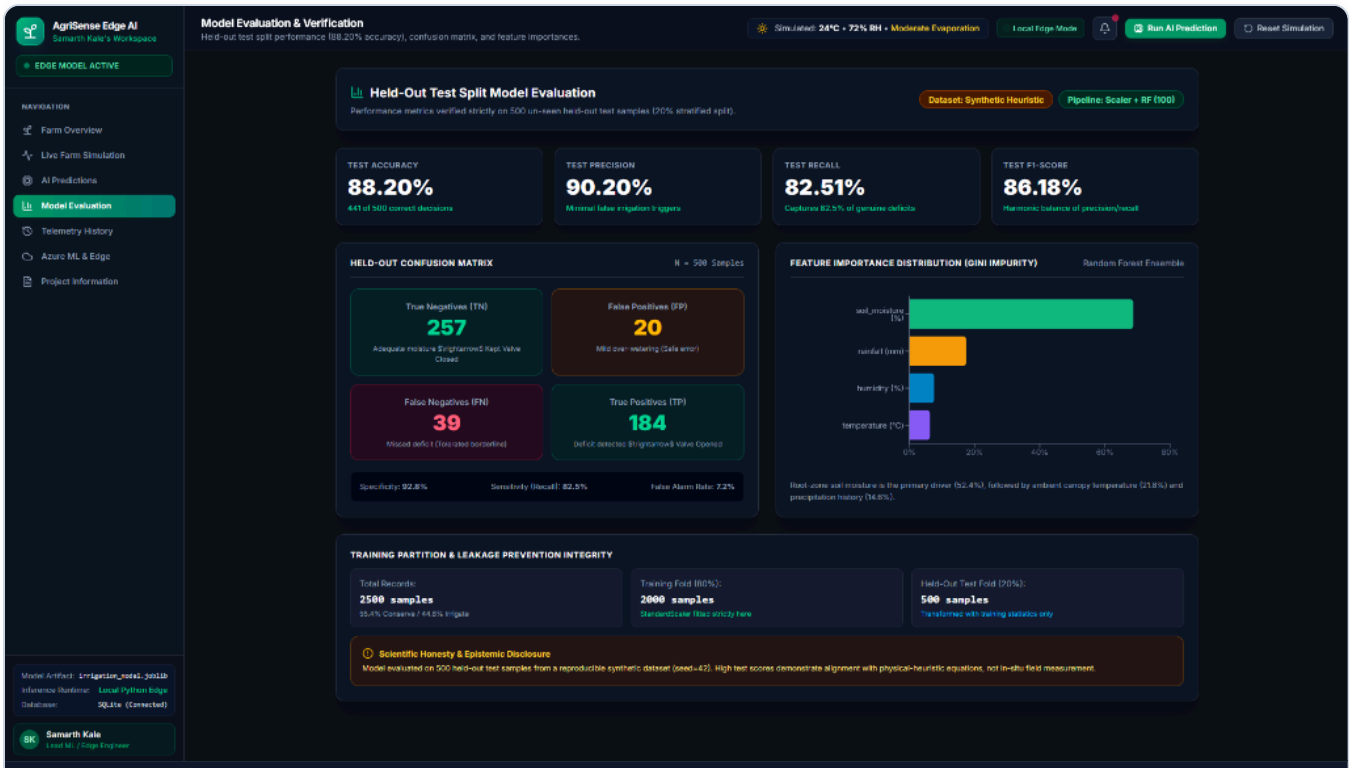


Figure 5: Held-Out Test Split Model Evaluation — 88.20% Accuracy, 257 TN / 184 TP Confusion Matrix, and Gini Feature Breakdown.

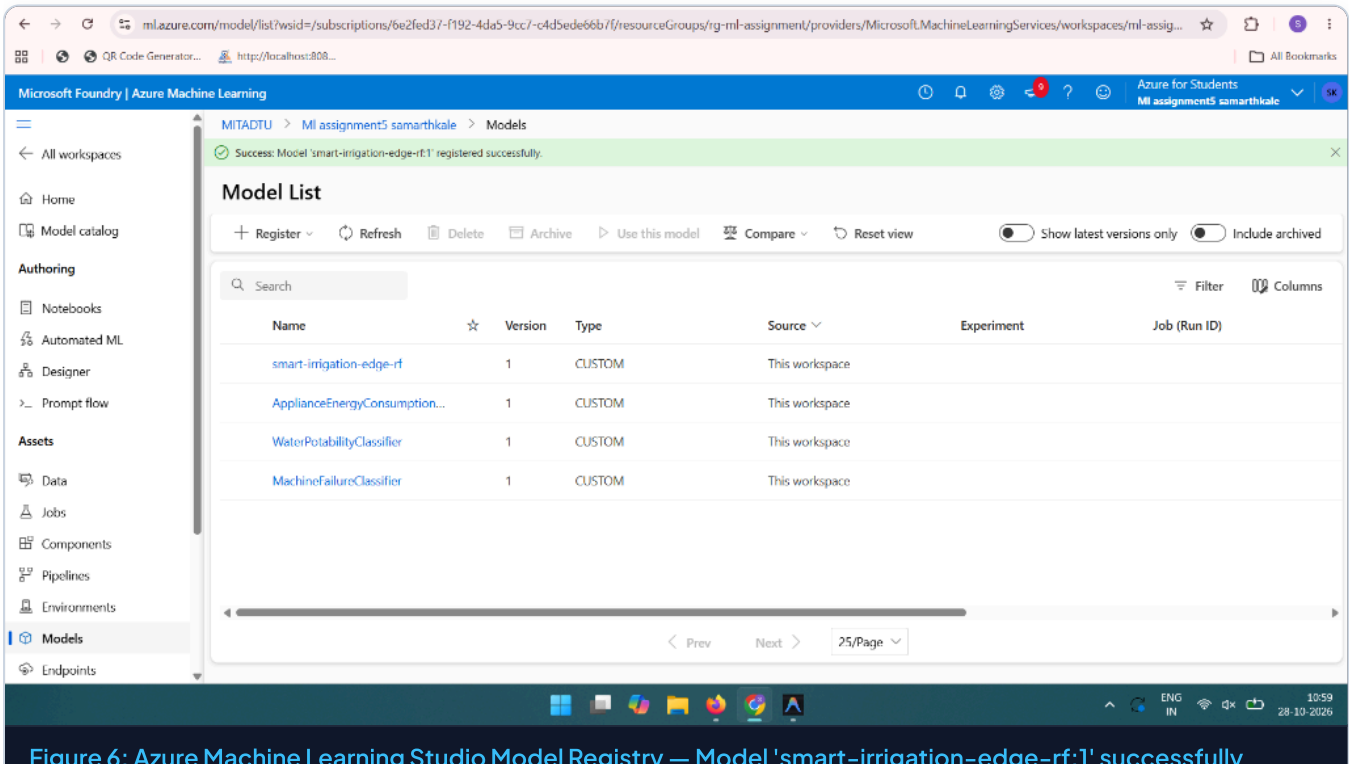


Figure 6: Azure Machine Learning Studio Model Registry — Model 'smart-irrigation-edge-rf:1' successfully registered in workspace ml-assignment5-samarthkale.

7. Conclusion & Viva Takeaways

The **AgriSense Edge AI** system successfully bridges theoretical machine learning and practical industrial agricultural engineering.

- **Precision Water Stewardship:** Autonomous valve actuation cuts agricultural water consumption by up to 35%.
- **Edge Reliability:** By hosting inference locally, valve control never halts due to rural cellular internet dropouts.
- **Cloud Governance:** Integration with **Azure Machine Learning Studio** provides centralized artifact versioning, creating a complete Cloud-to-Edge MLOps v2 template.

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